MIL-S-36439B
30 September 1976
SUPERSEDING
MIL-S-36439A
10 June 1974

MILITARY SPECIFICATION

STRETCHER, AMBULANCE, TWO-LEVEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of two-level ambulance stretcher, suitable for use in ambulances and station wagons.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-wall.

MILITARY

MIL-M-40024 - Mattress, Stretcher

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 S. 20th St., Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6530

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STANDARDS

FEDERAL

FED. TEST METHOD STD. 151 - Metals, Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified in the procurement document, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.1.2.1 and 6.2).

3.2 Material. Materials shall be as specified herein and shall be suitable in all respects for intended purpose.

3.2.1 Aluminum.

3.2.1.1 Tubing. Shall be high-tensile-strength, air-cylinder-quality aluminum tubing.

3.2.1.2 Rods and bars. Shall be high-tensile-strength aluminum.

3.2.1.3 Castings. Shall be high-tensile-strength, self-aging aluminum.

3.2.2 Helicals. Helicals shall be formed of automatic, coiler-grade, zinc- or cadmium-plated , spring-steel wire.

3.2.3 Hardware. Nuts, bolts, screws, roll pins, and washers shall be steel, zinc- or cadmium-plated .

3.2.4 Stampings. Shall be hot-rolled steel, zinc or cadmium-plated.

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3.3 Design. The ambulance stretcher shall be designed for use at two levels: low, for use in the ambulance; and high, hospital-bed height, for use in transferring the patient. The stretcher shall be capable of being raised and lowered easily. The design shall include an automatic locking device to retain the stretcher in the raised position, and a manual release for lowering it. The locking device and manual release shall be designed to prevent the stretcher from being released inadvertently from the raised position.

3.4 Construction. The ambulance stretcher shall consist of a caster-mounted base frame, a raising and lowering mechanism, a bed frame, and a mattress pad. The stretcher shall be constructed as shown in figure 1. There shall be no distortion or permanent set to parts when tested as specified in 4.4.3. The stretcher shall roll smoothly and easily when tested as specified in 4.4.2.

3.4.1 Base frame. The base frame shall consist of a tubular frame, offset caster-mounting arms, casters, and appropriate fittings.

3.4.1.1 Tubular frame. The tubular frame shall be of 7/8-inch-outside diameter by .125 inch aluminum tubing.

3.4.1.2 Mounting arms. Each corner of the base frame shall be provided with an offset caster-mounting arm, extending beyond the tubular frame in the longitudinal direction. The mounting arm shall be either an extension of the side members of the tubular frame, or an aluminum casting securely mounted to the tubular frame. The ends of the mounting arms shall be provided with fittings for mounting the casters.

3.4.1.3 Casters. The stretcher shall be provided with four 4-inch-minimum-diameter, double ball-bearing wheels with plain-bearing axles, each in a double ball bearing, swivel-type fork. The wheels shall be made of rubber and the forks shall be of cast aluminum alloy. The black rubber tire, molded to an aluminum hub (non-demountable) shall have a minimum width of 1 1/4 inches with a 7/8-inch minimum width tread. Caster stems shall be drilled or grooved to accept a setscrew which shall be used to secure the caster to the caster-mounting arm. Two of the casters, mounted on diagonally opposite corners, shall be provided with friction-locking, foot-operated wheel locks.

3.4.1.4 Fittings. Rubber bumpers shall be provided wherever the body section contacts the base frame, to prevent metal-to-metal contact. In addition, fittings shall be provided for location and attachment of the raising and lowering mechanism.

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KEY TO FIGURE 1

(1) Tubular base frame.
(2) Mounting Arms.
(3) Casters.
(4) Rubber bumpers.
(5) Securing clips.
(6) Uprights.
(7) Longitudinal cross members.
(8) Steel stampings.
(9) Transverse cross members.
(10) Lead handles.
(11) Foot-activated release mechanism.
(12) Body section.
(13) Tubular supporting structure and locking posts.
(14) Offset cross braces.
(15) Head section.
(16) Adjustment mechanism.
(17) Foot rest.
(18) Telescoping lift section
(19) Side rails.
(20) Spring-loaded locking device.
(21) Mattress.
(22) Heavy duty coil springs.
(23) Foot-operated wheel locks.

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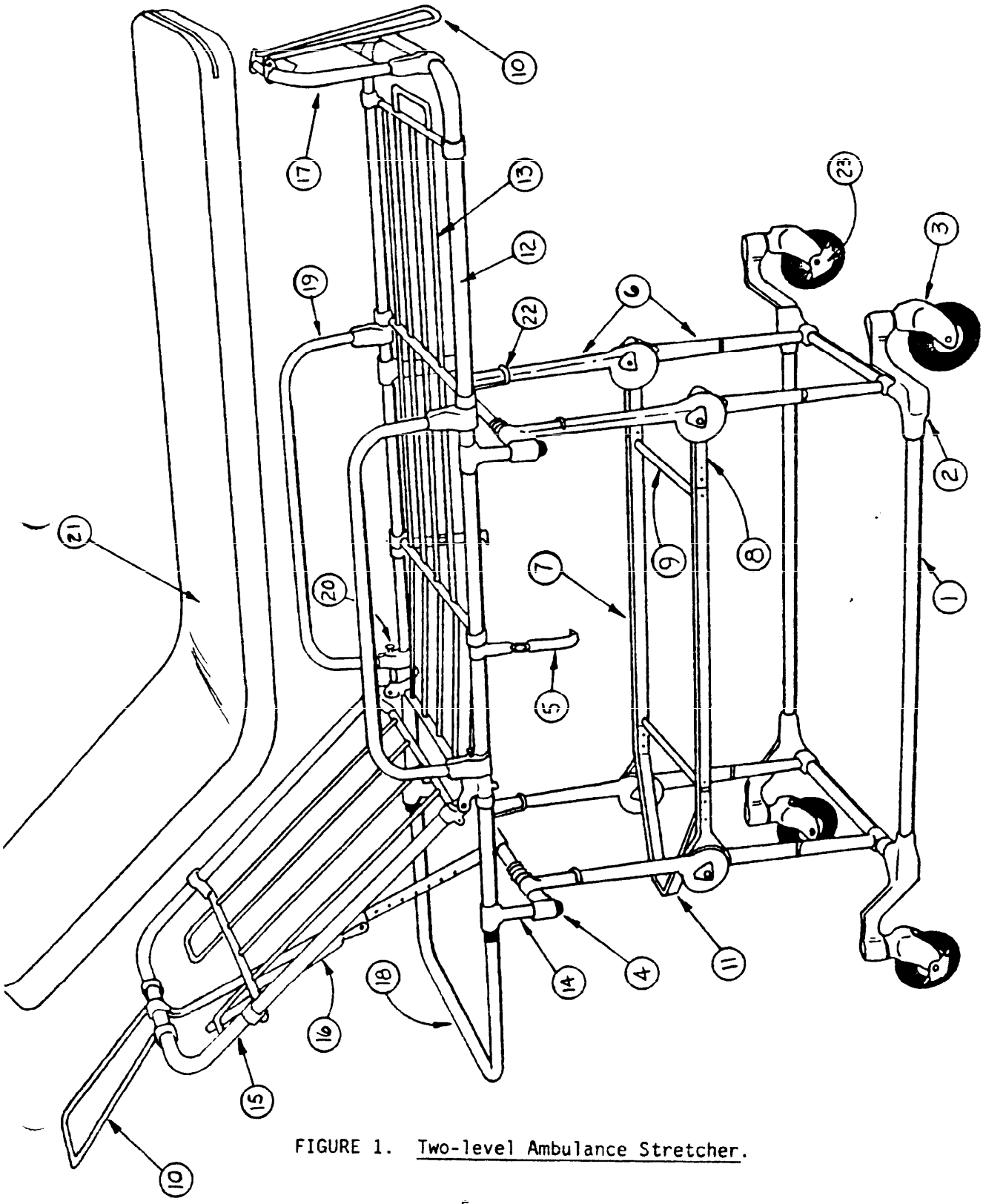


FIGURE 1. Two-level Ambulance Stretcher.

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3.4.2 Raising and lowering mechanism. The raising and lowering mechanism shall consist of a two-piece cast aluminum upright at each corner, connecting the base frame and the body-section; and cross-members joining the uprights. Longitudinal members and foot-activated release mechanism shall be of 1/4 inch by 1 inch rectangular aluminum bar. Steel stampings, designed to fit with the cast uprights, shall be connected to the longitudinal members. The transverse cross-members shall be 3/4 inch outside diameter by .083 inch thick (No. 14 Birmingham Wire (B.W.) gage) aluminum tubing. At the head and foot ends, the joint of the cast uprights and the steel stamping section of the longitudinal member shall be secured by means of a 1-3/4 inches long steel bolt, not less than 5/16-inch in diameter, and a locknut or a nut with lockwashers. Heavy duty coil springs shall be furnished and shall act to cause the vertical uprights to lock into the upright position.

3.4.3 Bed frame. The bed frame shall consist of a body section, an adjustable head section, a foot rest, a telescoping lift section, side rails, securing clips, and a lead handle at the head and foot of the stretcher.

3.4.3.1 Body section. The body section shall consist of a 1-3/32-inch-outside-diameter by .083-inch-thick (No. 14, B.W. gage) aluminum tubing U-shaped frame, and a tubular supporting structure. The tubular supporting structure shall have cross members made of 3/4-inch outside diameter by .083-inch-thick (No. 14, B.W. gage) aluminum tubing, and transverse members made of .405-inch outside diameter by .065-inch-thick (No. 16, B.W. gage) aluminum tubing. The U-shaped frame shall have corners bent on a 3 3/4-inch centerline radius. Two offset cross braces, made of 7/8-inch-outside-diameter by .083-inch-thick (No. 14, B.W. gage) aluminum tubing and appropriate cast aluminum fittings, shall support the body section and provide the connection for the upper ends of the vertical uprights. Locking posts for securing the stretcher in the ambulance shall be located on the underside of the body section. Posts shall be integral with the offset cross braces and shall be spaced 32 inches, center-to-center.

3.4.3.2 Adjustable head section. The adjustable head section shall be hinged to the body section, and shall consist of tubing frame and tubular supporting structure as described in 3.4.3.1. It shall have a sturdy adjustment mechanism which shall provide adjustments in the angle of the head section from horizontal to vertical with not less than seven intermediate positions. This full range of adjustments shall be possible when tested as specified in 4.4.2.

3.4.3.3 Foot rest. A tubular aluminum foot rest shall be designed for attachment to the foot end of the body section tubing. It shall be 11-1/4 inches wide by 8-3/4 inches high with a center pad section 5-1/4 inches high.

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3.4.3.4 Telescoping lift section. The telescoping lift section shall be one length of 7/8-inch diameter by .125 inch thick aluminum tubing, with 3-3/8 inch centerline radius bend at each corner. The lift section shall be provided with bearings to assure ease and freedom of telescoping and with means to prevent it from being removed from the bed frame tubing. There shall be no distortion causing failure of the telescoping function when tested as specified in 4.4.3.

3.4.3.5 Side rails. The side rails shall be made of 3/4-inch-outside-diameter by .083-inch-thick (No. 14, B.W. gage) aluminum tubing, fitted into cast aluminum fittings. The rails shall be attached to the tubular bed frame and shall rotate about it. The side rails shall lock in the vertical position by means of an automatic, spring-loaded device, provided with a release knob.

3.4.3.6 Securing clips. One securing clip shall be attached to each side of the tubular bed frame by means of a ring assembly. The clip shall be formed to fit over the longitudinal tubular member of the base frame, and shall hold the bed frame securely to the base frame when the stretcher is in the lowered position. The clip shall be coated with a plastic material, such as plastisol, to prevent scratching the surfaces of the base frame.

3.4.3.7 Lead handles. The lead handles shall be formed of .405 inch outside diameter by .065 inch thick (No. 16, B.W. gage) aluminum tubing. Handles shall be free-swiveling, one attached to the foot rest, and one attached to the head section of the stretcher.

3.4.4 Mattress. The mattress shall conform to the requirements contained in MIL-M-40024 as modified for the applicable National Stock Number (see 6.5).

3.5 Dimensions. The stretcher shall have the following overall dimensions:

Length	75 inches ± 1/2 inch
Width (bed frame)	22 inches ± 1/2 inch
Width (overall, including wheel locks)	24-1/4 inches ± 1/2 inch
Height of bed frame:	
Low position	9 3/4 inches maximum
High position	30 to 31 inches

3.6 Weight. The weight of the stretcher, without the matress, shall not exceed 55 pounds.

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3.7 Finish. All aluminum tubing and castings shall be given a satin finish or brighter. With exception of the tubing in the telescoping lift section and the hardware, steel parts shall be plated with cadmium or zinc.

3.8 Identification marking. The stretcher shall be marked, permanently and legibly, with the contract or purchase order number and the name or registered trademark of manufacturer.

3.9 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for operation, maintenance, cleaning, and lists of component parts.

3.10 Workmanship. Workmanship shall be first class throughout. Stretcher shall be free from defects which detract from its appearance or may impair its serviceability. All burrs and sharp edges shall be removed from the stretcher. Finish shall be uniform, free of scratches, discoloration and rough grind marks.

3.10.1 Welding. Welding shall be durable, sound in every detail and in accordance with the latest practices recommended by the American Welding Society. All welds shall be produced by qualified welding operators with satisfactory welding equipment using satisfactory filler metal or electrode where applicable to provide the best possible weld. Surfaces to be welded shall be free from grease, oil, oxides or any other foreign matter. Any cleaning method which will remove these substances and leave the surface clean is satisfactory. All weld joints shall blend into the adjacent metal in gradual, smooth curves. Beads shall be smooth and free of slag, excessive undercut, or excessive splatter. Sufficient metal shall be added to provide a suitable fillet or reinforcement unless otherwise specified. Welds shall not be oxidized. All welds shall be continuous with no visible voids, pin holes, or cracks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection.
b. Quality conformance inspection.

4.1.2.1.1 First article inspection. First article inspection shall be performed on the stretcher when a first article sample is required (see 3.1). This inspection shall include examination of 4.3. and tests of 4.4.

4.1.2.1.2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.3 and tests of 4.4.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and table I. The unit of product for sampling purposes shall be one stretcher.

TABLE I. Sampling for examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-1	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II, for an acceptance number of zero.

TABLE II. Sampling for tests.

Component (Unit of product)	Characteristic	Requirement	Test procedure	Inspection level
Raw material	Chemical composition	3.2	4.4.1	1/
Stretcher	Maneuverability	3.4, 3.4.3.2	4.4.2	S-2
	Load-bearing capability	3.4	4.4.3	S-2

1/ Raw materials. A sufficient quantity of each lot of each metal used in the manufacture of the stretchers shall be selected for chemical analysis.

4.3 Examination. The stretchers shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in table III. Examination shall not be restricted to the classified possible defects listed below.

TABLE III. Classification of defects.

Category	Defect
Major	
101	Design and construction not as specified.
102	Any part missing, damaged or otherwise unusable.
103	Any component not properly aligned or fitted.
104	Any weld incomplete, broken, or cracked.
105	Burrs or sharpness on exposed surface, edge, or corner.
Minor	
201	Finish not as specified, not uniform, not free of scratches, discoloration or rough grind marks.
202	Identification marking not present, permanent, complete, and legible.
203	Service data not supplied in duplicate.
204	Stretcher not clean, shows oil, dirt, or grease.
205	Weld not continuous, not clean, has pitting, spatter or discoloration.
206	Exposed welds not ground smooth to blend with adjacent surfaces.

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4.3.2 Dimensional examination. The stretchers shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.3.3 Packaging inspection. The inspection of the packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.4.1 Chemical composition. Four ounces from each lot of metal shall be tested in accordance with FED-TEST-METHOD STD 151.

4.4.2 Maneuverability. Six hundred pounds shall be distributed over a 6-foot section of the stretcher, in the raised position, 400 pounds to be placed in the body section and 200 pounds to be placed on the head section. With this load in place, the stretcher shall be rolled on a smooth surface in all directions on the casters. This test shall be repeated with the head section adjusted in all possible positions.

4.4.3 Load-bearing capability. The stretcher, in the raised position, shall be supported by means of horses at the extreme end of the foot section frame and at the crossbar of the telescoping frame, so that casters are off the floor. The frame shall be in the fully-extended position. The head section shall be in a horizontal position. The stretcher shall be loaded as specified in 4.4.2. Load shall be applied for a minimum of 1 hour. At the conclusion of the test, the telescoping frames shall be tested for freedom and ease of motion.

5. PACKAGING

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Unit of issue. One stretcher, with mattress, as specified, constitutes one unit of issue.

5.1.1.1 Level A.

5.1.1.1.1 Unit preparation. Each unit shall be disassembled as far as is practical. Casters shall be removed. Mattress shall be individually packaged in a bag fabricated of polyethylene film of .004-inch thickness with no wall thickness falling below .00325-inch. All seams and closure shall be effected by a continuous heat seal. Excess air shall be removed from bag before complete closure is effected. All exterior finished surfaces of the major item and disassembled parts shall be protected by wrapping with

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Kraft paper or polyethylene and cushioning with flexible corrugated fiberboard or other suitable materials. Wrapping and cushioning material shall be secured by tying with cotton cord or tape, or by sealing with Kraft gummed paper tape. Work shall be accomplished in such a manner as to insure against marring, scratching, or damage to the unit.

5.1.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Each unit shall be overpacked in an exterior container constructed in accordance with PPP-B-640, class 2, style E, regular slotted. The unit shall be immobilized within the exterior container in such a manner as to insure against shifting and breakage or marring of surfaces. Closure and sealing shall be in accordance with the appendix of the box specification. The use of steel strapping is not permitted.

5.2.2 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior package. Each interior package shall be identified as to contents.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The stretcher is intended for use as a wheeled stretcher in ambulances and station wagons, and to transfer patients from ambulances to hospital beds.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. When a first article is required (see 3.1, 4.1.2.1 and 6.3).

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6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of Armed Services Procurement Regulation. The contract or purchase order specifies the quantity, procedure for delivery and other specific instructions for inspection and approval of the first article.

6.4 This specification covers the following item(s) appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6530-00-982-2042	STRETCHER, AMBULANCE, Two Level

6.5 Mattress. The mattress referenced in 3.4.4 is covered under NSN 6530-00-784-2120. The correct requirements for this mattress shall be obtained from the Defense Personnel Support Center, 2800 So. 20th St., Philadelphia, Pa. 19101 ATTN: DPSC-ATT.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6530-1395

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-S-36439B STRETCHER, AMBULANCE, TWO-LEVEL

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
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☐ YES ☐ NO *(If "Yes", in what way?)*

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